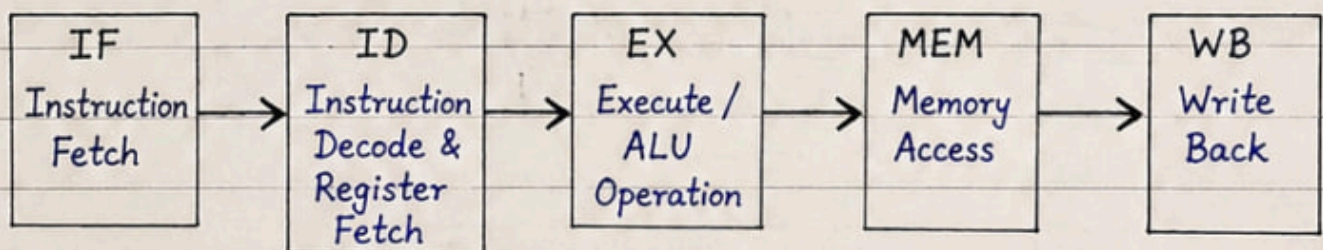


1. Definition: Pipeline architecture is a technique in which instruction execution is divided into a number of smaller stages. Different stages are executed simultaneously for different instructions using separate hardware units.

2. Working Principle:

- The instruction execution is divided into a sequence of stages.
- Each stage performs a part of the instruction.
- Multiple instructions are overlapped in execution in different stages.
- After the pipeline is full, one instruction completes in every clock cycle.

Example: 5-Stage Instruction Pipeline



Clock Cycle →	1	2	3	4	5	6	7	8	9
Instruction 1	IF	ID	EX	MEM	WB				
Instruction 2		IF	ID	EX	MEM	WB			
Instruction 3			IF	ID	EX	MEM	WB		
Instruction 4				IF	ID	EX	MEM	WB	
Instruction 5					IF	ID	EX	MEM	WB

3. Benefits:

- Increases throughput of the CPU.
- Higher CPU utilization.
- Improves performance.
- Better speedup with multiple instructions.

4. Limitations:

- Does not reduce latency of a single instruction.
- Pipeline hazards (data, control, structural) reduce efficiency.
- More hardware complexity.
- Performance decreases if stages are unbalanced.

Q. Pipeline Hazards: definition, types (Structural, Data, Control) & explanation.

Ans:

Definition: A hazard in pipelining is a condition that prevents the next instruction in the pipeline from executing during its designated clock cycle.

Types of Pipeline Hazards:

There are three main types of pipeline hazards:

1. Structural Hazards

- Occur when hardware resources are insufficient for the pipeline stages.
- Two or more stages need the same resource in the same clock cycle.
- **Example:** Single memory used for both instruction fetch and data access.

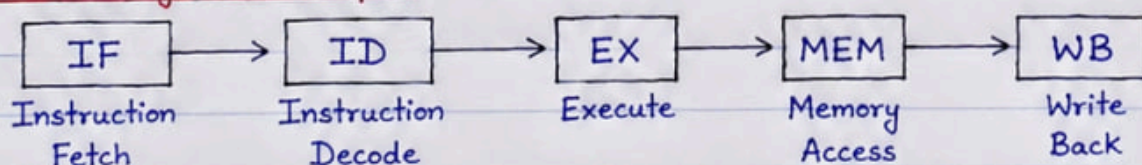
2. Data Hazards

- Occur due to data dependencies between instructions.
- An instruction depends on the result of a previous instruction still in the pipeline.
- **Types:** RAW (Read After Write), WAR (Write After Read), WAW (Write After Write).
- **Handled by:** Forwarding, stalling (bubbles).

3. Control Hazards

- Occur due to change in the flow of control.
- Caused by branch, jump or other control instructions.
- Next instruction address is unknown until the branch is resolved.
- **Handled by:** Branch prediction, stalling, flushing.

Pipeline Stages (Example):



Hazards reduce the efficiency of pipelining. They must be detected and handled to maintain pipeline performance.